



Development of a Generalized Deictic Framing Repertoire in an Autistic Child

Amanda N. Chastain¹ · Mark R. Dixon¹ 

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Abstract

The current study evaluated the effectiveness of relational training on the establishment of three deictic framing repertoires (I/You, Now/Then, Here/There) in an eight-year-old autistic boy using a multiple baseline across behaviors design. Relational training was effective in establishing all three directly trained and mutually entailed deictic relations, while differences were observed in transformation of stimulus function. Results support previous findings with the systematic development of multiple types of perspective taking repertoires with the same participant.

Keywords Verbal behavior · Relational Frame Theory · Deictic framing · Autism

Prior research has found that children diagnosed with autism spectrum disorder (ASD) show greater difficulty in perspective taking tasks relative to allistic children (Baron-Cohen et al., 1985; Rehfeldt et al., 2007). Further, impaired perspective-taking has been positively correlated with performance on measures of social skills (Dawson & Fernald, 1987). Perspective taking has been behaviorally described as an example of deictic relational framing and has been established using relational training in allistic (e.g., Heagle & Rehfeldt, 2006; Rehfeldt et al., 2007; Weil et al. 2011) and autistic children (e.g., Barron et al., 2019; Belisle et al., 2016; Gilroy et al., 2015; Jackson et al., 2014; Lovett & Rehfeldt, 2014). Moreover, improvements in complex (i.e., reversed and double-reversed) I/You, Here/There, and Now/Then deictic relations have been linked to improvements in

participant responses on traditional measures of perspective-taking (Weil et al., 2011).

In a recent example, Belisle et al. (2016) targeted the establishment of single-reversal I/You relations for three autistic children using a procedure adapted from the Promoting the Emergence of Advanced Knowledge System Transformation Module (PEAK-T; Dixon, 2016). During relational training, a card with images on both sides was positioned between the participant and researcher such that each could see a different image. Participants were asked to vocally identify the image facing them in the context of a single-reversal “You” relation (e.g., “If I were you and you were me, what would you see?”) and multiple probes for the mutually entailed single-reversal “I” relation (e.g., “If I were you and you were me, what would I see?”). All participants exhibited mastery of the single-reversal I/You relations, and two of the three correctly responded to tests for mutual entailment and transformation of stimulus function to an untrained stimulus set. One participant required remedial training involving mixed trial training blocks containing single-reversal I and single-reversal You relations, following which they also demonstrated the emergence of correct responses to the single-reversal I/You relations with an untrained stimulus set. Another recent example was that of Barron et al. (2019) who targeted the development of Here/There and Then/Later relations within participants using a multiple baseline across programs. In their analysis, relational training was effective for teaching two six-year-old children diagnosed with autism to correctly respond to

- A practical demonstration of the use of relational training in an autistic child.
- The establishment of three distinct generalized deictic framing repertoires (I/You, Here/There, Now/Then).
- Use of multiple exemplar training to yield positive outcomes for transformation of stimulus function via relational training.
- Identifies evidence-based teaching methodologies for the establishment of perspective taking repertoires.

✉ Mark R. Dixon
mrdixon@uic.edu

¹ Department of Disability and Human Development, College of Applied Health Science, University of Illinois Chicago, Chicago, IL 60608, USA

both single reversal Then/Later and Here/There relations. They additionally demonstrated success with transformation of stimulus function of both skills in response to untrained stimuli. This type of analysis allows for the evaluation of the development of the various types of deictic relational operants for a single participant as opposed to analyzing data across multiple participants with differing histories of reinforcement and existing verbal behavior repertoires. No studies to date analyze the independent development of I/You, Here/There, and Now/Then relations within a single individual, and doing so may be helpful in understanding the functional independence or dependence on these semi-related perspective-taking skills. The present study used a multiple baseline design across behaviors to assess the systematic development of generalized deictic framing repertoire across I/You, Here/There, and Now/Then relational operants and to evaluate the effectiveness of relational training on the establishment of these framing skills.

Method

Participant, Setting, and Materials

The participant was an eight-year-old autistic white male attending a summer social group at an ABA clinic at a large midwestern university. Experimental sessions took place in a large room containing a conference table, several chairs, a small round table, shelves, a large window, a projector, a projector screen, and stimuli for various social games. The participant sat in front of the experimenter at a table. A 13-inch MacBook Pro was set up next to the participant such that the camera would capture all audio along with the participant’s selection responses.

The PEAK Comprehensive Assessment (PCA; Dixon, 2019) was administered at the onset of the study to assess the participant’s language and cognition. The PCA contains 344 total test items across five subtests (i.e., Direct Training, Generalization, Equivalence, Transformation Receptive, and Transformation Expressive) evaluating the individuals’ verbal behavior. The PCA results indicated significant differences in both receptive (40% age-typical) and expressive (22% age-typical) deictic framing sub-domains relative to normative data.

Training procedures were adapted from the following three PEAK-T programs: DTC 7D- Now and Then (Past), DTC- 9P- Single Reversal Here/There, and DTC 13F- I/You Reversal (Dixon, 2016). Stimuli consisted of paper images measuring two inches in height by two inches in length. Stimuli used in each program are represented in Table 1. Stimuli for the Now/Then (Past) program consisted of six pairs of familiar images separated into two stimulus sets. The Now/Then (Past) program used six pairs of familiar images

Table 1 Visual representation of stimuli used across programs

Exemplar	Now/Then			
	Set 1	Set 2		
1	Apple/Sock	Car/Fork		
2	Flower/Shoe	Happy Face/Lollipop		
3	Dresser/Cup	Dog/Butterfly		
Exemplar	Here/There single reversal			
	Here stimuli	There stimuli		
1	Pen	Sand (Beach)		
2	Binder	Slide (Park)		
3	Crayons	Giraffe (Zoo)		
Exemplar	I/You single reversal			
	A stimuli	B stimuli	C stimuli	D stimuli
1	Corn	Tomato	Sally	Bob
2	Sun	Moon	Lax	Rem
3	Fox	Snail	Cam	Jeb

divided into two stimulus sets. The Here/There program used images representing items in the therapy room (Here) and other locations (There), along with pen and paper (4x5 inch) for drawing during transformation tests. The transformation test included novel objects and locations created by the participant. Finally, the I/You program utilized pairs of images on back-to-back cards, each linked to a pseudonym for the experimenter and participant. Pen and paper were provided for the I/You transformation task.

Dependent Measures

The dependent variable for this study was participant Response Score for each training and testing block. Participant responses were scored as outlined in the PEAK Curriculum, where a score of 10 indicates an independent correct response was emitted on the first attempt, a score of 8 indicates correct responding following the delivery of one prompt subsequent to a non-response or incorrect response, a score of 4 indicates correct responding following the delivery of two prompts subsequent to non-responses or incorrect responses, a score of 2 indicates correct responding following three prompts subsequent to non-responses or incorrect responses and/or a reduced stimulus array, and a score of 0 indicates no responses or incorrect responding after the delivery four or more prompts. The total response score was summed for each 10-trial block to obtain a total Response Score.

Procedure

A multiple baseline design across behaviors was used to evaluate the effect of relational training on participants trained and untrained responses across different deictic

relational operants (see Fig. 1 for training and testing relations). Initial baseline probes assessed all target relations. Training and testing conditions were introduced sequentially for Now/Then (past), Here/There, and I/You relations. After mastery of Now/Then relations, baseline probes were re-administered for Here/There and I/You relations, respectively. A third baseline probe was conducted for I/You relations following mastery of the Here/There program.

Each block included 10 trials, except for the Single Reversal Here/There program, which had five trials due to its complexity. Target stimuli were presented semi-randomly, with the same target stimulus presented for no more than two consecutive trials. A token economy system incorporated praise and points contingent on correct responses during training. Tokens were given for readiness skills and basic imitation to maintain engagement, and the backup reinforcer was given at the end of each test block. Differential reinforcement was implemented such that a prompted or independent response initially resulted in praise and a point, regardless of prior errors during that trial. Once a correct and independent response was emitted on the first attempt (i.e., a score

of 10), this was required to receive a point for that target, while scoring a 2, 4, or 8 resulted only in neutral praise. During training trials, points were exchanged for backup reinforcers after accumulating 5–10 tokens, as negotiated at the start of each token board. All test trials were scored as a 0 or 10 and resulted in neutral feedback. The participant was allotted up to 5 s to initiate a response during all trials. Relational training was considered mastered when the participant achieved a score of 90 or above for three consecutive training blocks. Blocks of testing, including transformation tests, were interspersed between training blocks as outlined in the subsequent sections below.

If mastery was reached without an increase in transformation test scores over three consecutive blocks, exemplar training was introduced. This involved training one stimulus class (exemplar 1) from the test sets using the same methodology outlined above. Following mastery of exemplar training, a transformation test probe was conducted with the remaining untrained stimuli. If necessary, this process would continue with additional exemplars until mastery was achieved.

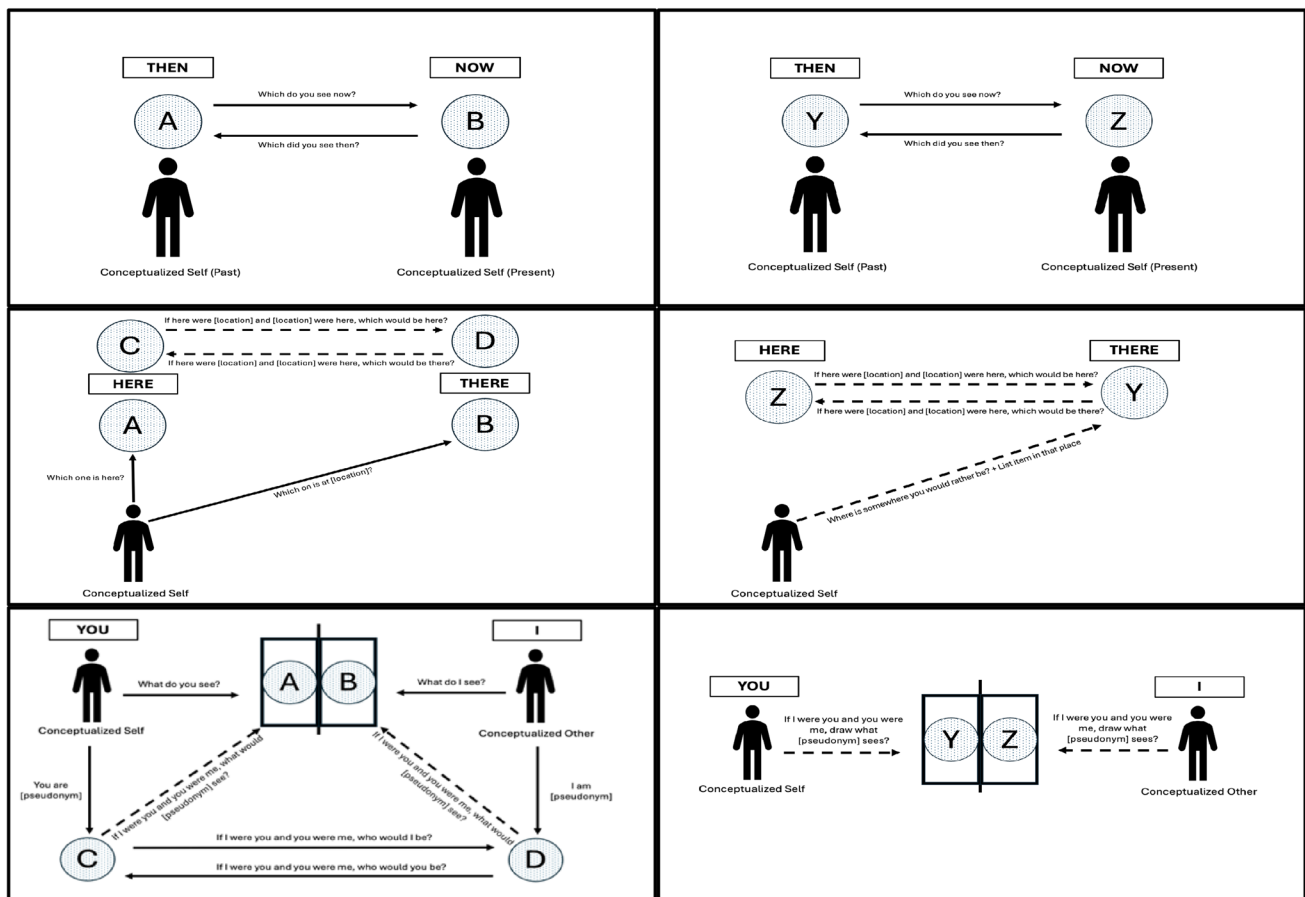


Fig. 1 Diagram of training and testing relations. Training relations are depicted with the solid arrows and test relations are depicted with broken arrows

Procedural fidelity was calculated for a minimum of 33% of trials across all conditions. A trial was scored as correct if correct sample and comparison stimuli were presented as applicable, the correct instruction was given, and the correct consequence (i.e., neutral feedback, praise, points, error statement) was delivered within 2 s of the participants' response. A trial was scored as incorrect if any of these conditions were not met on a given trial. The total procedural fidelity for each condition was calculated using the formula $\{[\text{total trials scored as correct} / \text{total number of trials}] \times 100\}$ and was observed to be above 80% across all training and testing blocks. Interobserver agreement was calculated for a minimum of 33% of trials across all conditions using the formula for total agreement $\{[\text{total agreement} / \text{total disagreements}] \times 100\}$ and was reported to be above 80% across all training and testing blocks.

Baseline Probes During this phase, all target relations were tested, including Now/Then (A-B), I/You (A-B and C-D), Here/There (A-B, C-D, CA/DB), and transformation (Y and YZ) relations. Each relation was presented as outlined in the training and transformation sections below.

Now/Then Training and Testing (A-B and Y-Z) During blocks of Now/Then Training, the participant was taught to label images he saw *now* versus *then*. The first image in the stimulus pair was presented for 2 s. That image was then removed, followed by the presentation of the second image in the stimulus pair, which remained visible while the experimenter asked, "Which did you see then?" (A) and "Which do you see now?" (B). A response was considered correct when the participant vocally labeled the image that corresponded with the Now/Then relation. For example, when the participant was first shown the image of the apple followed by the image of the sock and asked, "What did you see then?" the correct response would be saying "Apple" and when asked "What do you see now?" the correct response would be "Sock." Order of stimulus presentation and corresponding instruction (i.e., now vs. then) were varied through a trial block. Stimulus set 1 was used in this condition. Stimulus set 2 was used to test for transformation of function (YZ) to novel stimuli.

Here/There Training and Testing (A-B, C-D, Y, Y-Z) During blocks of Here/There Training the participant was taught to identify where objects would be found if locations were reversed. During A-B training blocks, the participant was presented with two images in a stimulus pair and asked to vocally identify which item was here and which was in another location when given the instructions, "Which one is here?" and "Which one is at [location]?" respectively. Each trial included both Here and There instructions, and a response was scored as correct if the participant correctly labeled the image corresponding with the Here/There

relation. For example, when the participant was presented with images of a pen and sand followed by the instruction "Which one is at the beach" the correct response would be selecting the picture of sand, and when asked "Which one is here" the correct response would be selecting the picture of the pen. Both questions were presented in a single trial.

After each A-B training block, the participant was taught the corresponding C-D relation, which involved identifying where objects would be if locations were reversed. For instance, when the participant was presented with the images of a pen and sand followed by the instruction, "If here were the beach and the beach were here, what would be here?" the correct response would be to say "sand," and the correct response to the question "If here were the beach and the beach were here, what would be there?" would be to say "pen." As with the no reversal condition, both here and there questions were asked in a single trial and the order of stimulus presentation and corresponding instruction (i.e., here vs. there) were varied through a trial block.

Transformation of stimulus function for Here/There involved two parts. In part 1 (Y), the participant was asked the the question, "Where is somewhere that you would rather be?" and subsequently asked to write the name of something they would see in that location. For example, if the participant said he would rather be at the grocery store, he may write down "milk" on the paper. Next, the participant was also to write the name of an object that was "here" (i.e., in the room), to which he may respond by writing the name of any item in the room. After writing an object for both here and there for a given trial, the YZ task was presented. YZ trials were procedurally identical to the C-D relations described above and used stimuli generated during the preceding Y trial.

I/You Training and Testing (A-B, C-D, C-A, D-B, Y-Z) In A-B training, the participant was first shown both sides of a picture card with images on opposing sides. He was subsequently shown one side of the card and asked, "What do you see?" and "What do I see?" and asked to name the corresponding image. Both instructions were present on each trial, and a response was scored as correct if the participant named the image that corresponding with the I/You relation. For example, the participant was shown a card with an image of a fox on the side facing them and a snail on the side facing the experimenter and given the instructions, "What do I see?" and "What do you see?" He responded by saying "Fox" and "Snail," respectfully.

During C-D training blocks, the participant was taught reverse roles by answering the questions, "I am [experimenter pseudonym] and you are [participant pseudonym]. If I were you and you were me, who would I be?" and "Who would you be?" These pseudonym pairs were pre-assigned to a given stimulus class. For example, the participant was

asked “I am Sally, and you are Bob. If I were you and you were me, who would I be?” where a correct response was stating the pseudonym “Bob.” In the same trial, the participant was then asked, “Who would you be?” where a correct response was stating the pseudonym “Sally.” As with the no-reversal condition, both I and you questions were asked in a single trial and the order of stimulus presentation and corresponding instruction (i.e., I vs. You) were varied through a trial block. The programmed consequences were the same as described in the Now/Then Training condition above.

C-A and D-B test trials were also presented to evaluate single-reversal I/You relations. The participant was presented with the same visual images used in the no reversal I/You condition and asked, “If I were you and you were me, what would [pseudonym corresponding with I] see?” and “What would [pseudonym corresponding with you] see?” The presentation of images was otherwise identical to the no reversal I/You condition, and programmed consequences were identical to baseline during testing. If direct training was required, programmed consequences were identical to the training outlined above.

During the transformation test for I/You, the participant was presented with a piece of paper and a pen, as well as the picture card described in I/You training. He was asked “If I were you and you were me, draw what [experimenter pseudonym] sees,” and “Draw what [participant pseudonym] sees.” Both I and You questions were asked in a single trial and the order of stimulus presentation and corresponding instruction (i.e., I vs. You) were varied through a trial block. For example, the participant was presented with a card showing the image of a fox on one side and a snail on the other, with the fox facing them. Then, the participant was asked, “If I were you and you were me, draw what Job sees,” and following a response or 5 s without beginning to draw, the participant was then asked, “Now draw what Cam sees.”

Results and Discussion

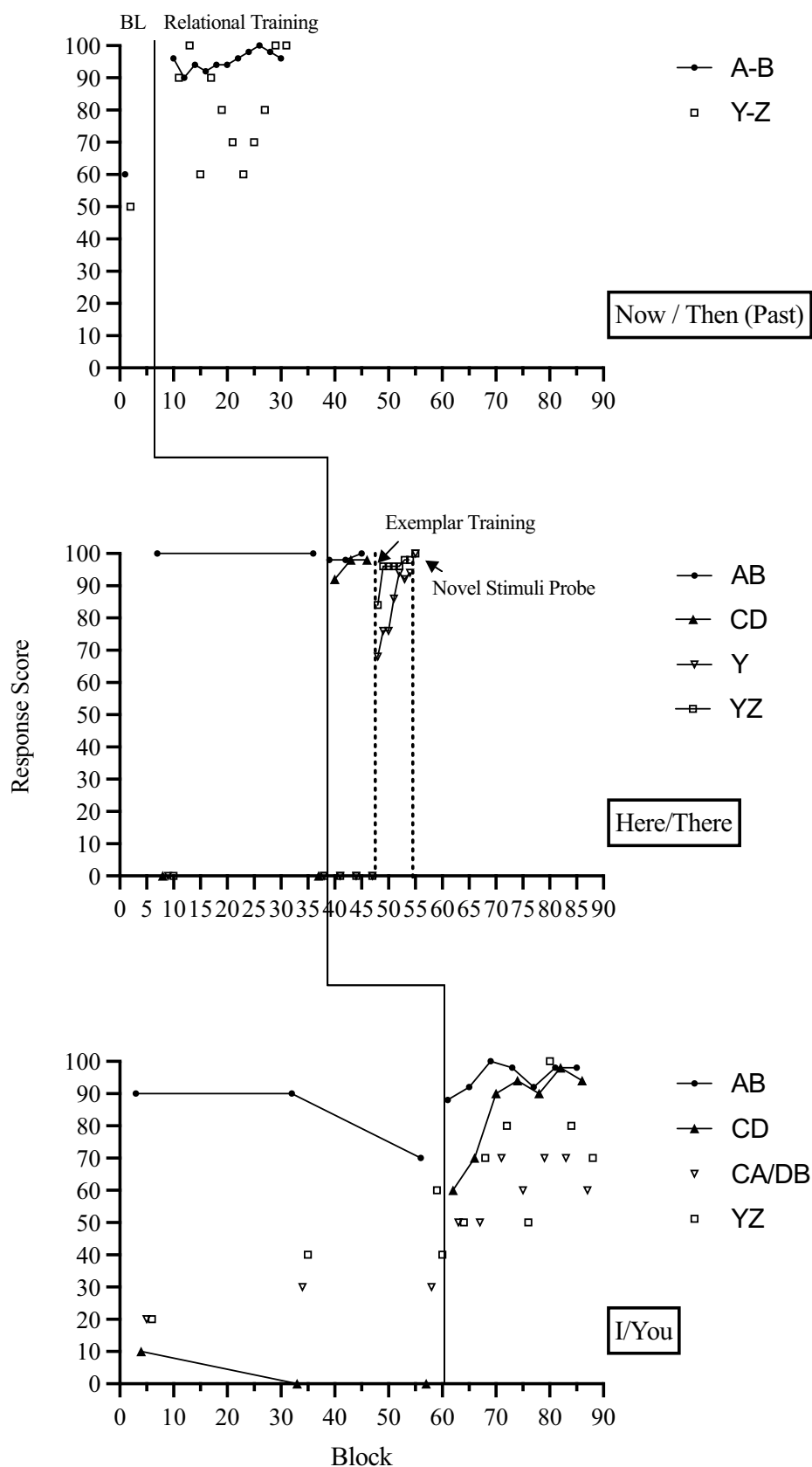
Figure 2 depicts the participant’s scores across programs for training and testing relations. During the initial baseline probes, performance was above criteria for Here/There A-B and I/You A-B relations, with response scores of 100 and 90, respectively. This was expected given the participant’s PCA scores. All other initial baseline scores were below mastery criteria, with response scores ranging from 0–60. After initial baseline probes, Now/Then training was introduced, with Y-Z probes following each A-B training block. Performance improved immediately for both training (90–100) and testing (60–100). A-B mastery was reached within three blocks (90–96). Owing to the variability in Y-Z scores, A-B training continued until mastery was achieved

after 22 blocks (11 A-B and 11 Y-Z). After mastering Now/Then relations, additional baseline probes for Here/There and I/You relations were conducted, consistent with a multiple baseline across behaviors design (c.f. Kazdin, 1982). The participant maintained a score of 100 on Here/There A-B relations, while C-D, Y, and Y-Z scores were all zero. For I/You relations, A-B scores remained high at 90, but C-D, CA/DB, and Y-Z scores were below mastery criteria, at zero, 30, and 40, respectively. Mastery of Here/There A-B (98–100) and C-D (92–98) relations was achieved after three blocks, while Y and Y-Z scores remained at zero, resulting in exemplar training. The participant reached mastery criteria after one round of exemplar training, consisting of seven blocks of Y (68–94) and Y-Z (84–98) relations. Subsequent transformation tests with untrained targets yielded scores of 100 for both Y and Y-Z relations. A baseline probe was then conducted for I/You relations, yielding scores of 70, 0, 30, and 60 for A-B, C-D, C-A/D-B, and Y-Z relations, respectively. Although Y-Z scores increased from 30 to 60 between probes two and three, a subsequent Y-Z probe showed no further improvement, with a score of 40. Mastery criteria was achieved after four blocks of A-B (88–100) and five blocks of C-D (60–98) training. While small gains were observed on tests for C-A/D-B (50–70) and Y-Z (50–100) relations, scores stabilized after seven blocks and mastery was not achieved. Exemplar training was not introduced owing to the end of the summer program.

This paper aims to provide a brief demonstration of the establishment of deictic framing repertoires as a result of relational training. Consistent with prior literature (e.g., Barron et al., 2019; Belisle et al., 2016; Gilroy et al., 2015; Lovett & Rehfeldt, 2014), relational training was effective in teaching Now/Then relations with the first set of stimuli, and transformation of stimulus function was demonstrated with untrained stimuli. Additionally, relational training was effective in teaching Here/There and I/You single reversal training relations; however, following mastery of A-B and C-D training, transformation of stimulus function was not observed for either deictic framing repertoire. It is possible that the participant’s history of reinforcement was not robust enough to result in contextual control over the novel responding involved in these transformation tasks. Therefore, exemplar training was introduced. Exemplar training of Y and Y-Z relations resulted in mastery and emergence of novel responding for Here/There deictic framing. Should replications of the current study yield similar results, researchers should implement exemplar training for I/You emergent relations (CA-DB and Y-Z) to determine its effectiveness on generative responding.

While the multiple baseline across behaviors design is useful in evaluating the effect of relational training on individual behavioral repertoires, the current analysis contains only one participant. Future research should replicate these

Fig. 2 Participant data. *Note.* BL indicates baseline probes. Trained relations are depicted using connected markers while untrained (tested) relations are depicted using disconnected markers



methods with additional participants to strengthen the generality of these results. Finally, this analysis lacks evidence of practical perspective-taking in everyday contexts, such as understanding others' perspectives, time, space, or emotion regulation. Future research should investigate generalization to real-world activities involving I/You, Here/There, and Now/Then relations, and examine key social skills such as empathy, inferencing, and predicting others' experiences. Incorporating probes for traditional perspective-taking measures after mastery of each deictic frame relation may also offer insights into the development of a comprehensive perspective-taking repertoire.

Availability of Data and Materials All data reported in this study are stored in the Cognition, Behavior, and Mindfulness Lab at the University of Illinois, Chicago. These data are available upon request from the first or second author.

Declarations

Conflict of Interest The second author declares small royalties from the sales of the PEAK curriculum. The first author declares no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent and assent were obtained from all individual participants in the study.

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